

**Course Week Focus: Week 12 (Cloud Deployment & Hosting) & Week 13
(Security Implementation)**

Contribution

As Project Manager, I coordinated the team's transition from local development to production. I organized two task breakdowns for Weeks 12 and 13, set internal deadlines, and facilitated quick decision-making when we encountered deployment errors. As GitHub Manager, I restructured the repository, protected the main branch, reviewed all pull requests, and ensured no secrets were committed. I also led the actual deployment to Render and the integration of MongoDB Atlas.

Actions & Group Experience

My Actions:

- Created the deployment checklist from the lab guide (Render + MongoDB Atlas).
- Configured environment variables securely on Render (e.g., DATABASE_URL, SESSION_SECRET).
- Managed the team's GitHub workflow: created a develop branch, enforced PR reviews, and tagged the production release.
- Resolved a "port already in use" conflict before deployment.
- Group Experience / Challenges:
 - Initially, the team struggled to differentiate between local vs cloud environments (e.g., localhost vs Render URL). I ran a 10-minute walkthrough to clarify.
 - One team member accidentally pushed .env to GitHub. I immediately rotated the exposed keys and added .env to .gitignore retroactively.
 - Outcome: By end of Week 12, the group successfully deployed a live full-stack app on Render, connected to MongoDB Atlas.

Week 13: Security Implementation – Actions & Group Experience

My Actions:

- Assigned security tasks based on the lab guide: input validation Migel. XSS prevention Jhed, rate limiting Glenn, password hashing Vince.
- I personally reviewed and merged all security-related PRs, testing each endpoint using Thunder Client.
- Enforced the use of helmet and express-rate-limit across all routes.
- Group Experience / Challenges:
- The team initially overlooked output encoding for XSS; we discovered a reflected XSS vulnerability during Thunder Client testing. I paused merges until all user-supplied data was sanitized.
- Rate limiting caused friction – legitimate requests were blocked during testing. We adjusted the windowMs and max limits as a group.
- Outcome: By end of Week 13, all API endpoints passed Thunder Client security tests. The team felt confident preventing brute force and XSS attacks.